Vector Database Operations and Maintenance

Maintaining vector databases in production environments involves continuous optimization and management. Document updates require reembedding changed content and updating corresponding vectors. Deletion removes vectors and associated metadata, either logically through soft deletion or physically removing entries. Batch operations process multiple vectors efficiently, crucial for high-volume updates. Incremental indexing accommodates continuous data ingestion without full reindexing. Index optimization rebuilds indexes periodically, improving search performance and memory efficiency. Vacuum operations reclaim space from deleted entries. Monitoring tracks index size, search latency, recall quality, and resource usage. Replication maintains multiple index copies for high availability. Sharding distributes vectors across multiple nodes, enabling horizontal scaling. Backup and recovery procedures protect against data loss. Migration between vector database products requires re-embedding all documents with new embedding models, a potentially expensive operation. Performance tuning adjusts parameters like number of neighbors examined during search, quantization levels, and memory allocation. Testing ensures consistency across replicas. Versioning tracks index versions enabling rollback if issues arise. Cost optimization balances query latency, accuracy, and infrastructure requirements. Understanding operational aspects is essential for reliable, scalable RAG systems.